# Agentic work. Economic clarity.

Vector brings cost visibility and workflow controls to model-driven work, with verification kept in the loop.

**5.0%**

**Lower recorded API cost**

Aggregate result across three controlled coding workflows.

**76/76**

**Specified checks passed**

Both runtimes passed every specified acceptance check in the set.

## Measured efficiency. Explicit quality gates.

The evaluation held the model and High reasoning constant. Under the fixed review rubric, Vector scored no lower on any of the three tasks. Recorded cost was $2.0209 for Vector and $2.1276 for Codex.

The Tokenomics Engine makes the price of work visible, applies configurable controls and supports disciplined improvements to the surrounding workflow.

Scope: small adaptive evaluation, one trial per task. Cost varied by workload; Vector was cheaper on one of three tasks. Results describe the tested candidate and measured gates. [E1]

# Measured on three workflows.

Same model. High reasoning. One autonomous agent per runtime. All recorded runtime requests retained; no evaluator-triggered repair in this set.

| Workflow | Vector API cost | Codex API cost | Checks each | Review /48 Vector / Codex |
|---|---|---|---|---|
| **Workflow Runner** | $0.7513 | $0.8920 | 24/24 | 45 / 45 |
| Evidence Workbench | $0.7277 | $0.7229 | 26/26 | 47 / 47 |
| Delivery Outbox | $0.5420 | $0.5127 | 26/26 | 47 / 46 |
| **Three-task total** | **$2.0209** | **$2.1276** | **76/76** | **139 / 138*** |

*Score totals are descriptive (maximum 144). Quality is checked on each task independently. Costs cover runtime API usage at study rates; development and independent evaluation are excluded. This is not a subscription-price or invoice comparison. [E1]

## Workflow Runner: a named case study

**15.8%**
**Lower recorded API cost**

**22.7%**
**Less agent wall time**

Both passed 24/24 checks and scored 45/48. This is one task in the complete table above; it does not predict every workflow.

Observed cost variation: Vector was 0.7% more expensive on Evidence Workbench and 5.7% more on Delivery Outbox. The aggregate reduction was 5.0%.

# Optimize the cost of work.

Vector's Tokenomics Engine connects execution, usage visibility and verification. Its purpose is to make useful work more accountable and avoid needless repetition.

**Define**
Objective and quality criteria

**Coordinate**
Model, tool and agent work

**Measure**
Usage and workflow cost

**Control**
Limits and work continuity

**Verify**
Checks and review evidence

## See the workflow

Bring primary, delegated and helper model usage into a clearer view of the work. Keep known charges distinct from incomplete or uncertain usage.

## Keep useful continuity

Support cache-aware context continuity and combine routine tool setup with readiness checks, while retaining permissions and relevant verification.

## Control repetition

Apply configurable run, step and loop controls. Preserve explicit user rechecks and useful progress as work continues.

### Raw token count is not the bill.

In the three-task set, Vector generated 16.8% less output while processing 28.3% more input. The weighted mix of output, cache writes and cache reads produced the lower total cost. [E2]

This is a functional overview. Spending estimates and after-step controls support visibility and limits; they are not a guaranteed hard invoice cap.

# Progress that can be audited.

A disciplined cycle connects measured behavior to product changes and regression checks.

**281**

**Accounting tests passed**

Eight baseline failures resolved. [E6]

**23**

**Package typechecks passed**

Completed-call accounting fixes. [E8]

## What the current controls address

Earlier 220 tests cover repeated outcomes, rechecks and continuity. Latest completed-call accounting passed 109 usage and 37 budget/abort tests, with eight existing skips. A 24-case failure matrix passed, including 12 cases that failed before the latest fix. Suites overlap and are not additive. API cost benefits from these accounting changes have not been isolated. [E3, E6, E9]

## Benchmark scope

Three custom synthetic task pairs on Windows used the same gpt-6.1-sol model, High reasoning and one autonomous agent per runtime, with one trial per task. Two tasks were development replications; one was held out. Results describe the tested candidate, required checks and fixed static rubric. The sample does not establish statistical significance or predict all workloads.

Prices are derived from recorded provider usage at the study schedule. Reviews use exact source and selected screenshots; passing checks do not certify production reliability or security. Large-team episodes are reported separately with their stopping conditions and quality limits.

### Evidence register

**E1** Three-task matched benchmark and quality gate

**E2** Receipt-based cost category projection

**E6** Summary-cost regression and before/after validation

**E7** Cancellation accounting validation | **E8** 23-package verification | **E9** Saving-error checks

**E3** Earlier V2 control-suite validation

# Multi-agent stress validation.

## Nested seven-role API, CLI and SDK build

Same gpt-6.1-sol model and High reasoning; 7 configured agents per runtime. All recorded charges retained and both episodes settled.

### Vector

**$6.6391**

53/54 final-source diagnostic; 279 provider requests

Blinded static reviews: 46/48 and 47/48; agents 7/7

### Codex

**$4.6034**

53/54 final-source diagnostic; 300 provider requests

Blinded static reviews: 43/48 and 42/48; agents 7/7

### Full quality criteria were not met in this run.

Shown checks are separate free final-source diagnostics; Vector's original submission check was not run. The remaining assertion expects 401 rather than 403 for the bootstrap credential; both products denied access. Vector stopped at a conservative request-reservation limit; Codex reached 300 requests during repair. Both original native audits require review. Vector scored higher in both blinded review totals, with two lower dimension scores in one review. These descriptive costs and reviews do not establish equal-quality savings. The smaller three-task result remains separate.

Study evidence and original outcomes

# Budget-limited maintenance.

## Upgrading an existing billing service

Same gpt-6.1-sol model, High reasoning and seven configured agents per runtime, using the shared remaining study budget. Both episodes ended at the reservation guard; all 189 charges are retained and settled.

### Vector

**$1.4211**

53/54 final-source diagnostic; 58 provider requests

Blinded static reviews: 45/48 and 45/48; agents 5/7

### Codex

**$2.3666**

53/54 first-submission checks; 131 provider requests

Blinded static reviews: 48/48 and 47/48; agents 7/7

### Full quality criteria were not met in this run.

Codex submitted 53/54; its interrupted repair left the source unchanged. Vector's original check was not run; a separate free final-source diagnostic passed 53/54. Both rejected an invalid key scope with 404 where the gate expected 400. Vector stopped after five implementation roles, before independent QA and the judge; its source reviews also scored lower. This trial cannot establish savings at matching quality or compare fully completed workflows. Original outcomes and static findings remain in the evidence.

Study evidence and original outcomes